

Пробная олимпиада – 2

Time allowed: 3 hours

1. Evaluate

$$\frac{1/2}{1 + \sqrt[2]{2}} + \frac{1/4}{1 + \sqrt[4]{2}} + \frac{1/8}{1 + \sqrt[8]{2}} + \frac{1/16}{1 + \sqrt[16]{2}} + \dots$$

2. Let $f: [0, 2\pi] \rightarrow \mathbb{R}$ be such that

$$\int_0^{2\pi} f(x) \cos(kx) dx = 1 \quad \text{for all } k = 1, 2, \dots, n \quad (n \geq 2).$$

Find the minimum of $\int_0^{2\pi} f^2(x) dx$ over all such functions f .

3. Suppose that $P(x) = a_1x + a_2x^2 + \dots + a_nx^n$ is a polynomial with integer coefficients, with a_1 odd. Suppose that $e^{P(x)} = b_0 + b_1x + b_2x^2 + \dots$ for all x . Prove that b_k is nonzero for all $k \geq 0$.

4. Let B be an $n \times n$ positive-definite matrix with real entries. Assume that there exists an $n \times n$ real matrix C such that

$$BC^2 - B^2C = E \quad \text{and} \quad (\operatorname{tr}(C^2) - \operatorname{tr}(CB)) \cdot \operatorname{tr}(B) = 2809.$$

Find the maximum possible value of n .

5. For which dimensions n does there exist a hyperplane in $\mathbb{R}^n$ that intersects all $(n-1)$ -dimensional closed faces of the n -dimensional cube but has no common points with the closed ball inscribed in this cube?